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## TRAILING EDGE COOLING CIRCUIT FOR TURBOMACHINE AIRFOIL


### Abstract

An airfoil includes a leading edge, a trailing edge, and a pressure side wall and a suction side wall each extending between the leading edge and the trailing edge. The pressure side wall and the suction side wall define a main body region and a trailing edge region. The main body region extends from the leading edge to the trailing edge region and defines an upstream plenum. The trailing edge region extends from the main body region to the trailing edge. The trailing edge region defines pressure side channels and suction side channels in fluid communication with the upstream plenum and arranged in a trussed structure to reduce stress in the trailing edge region. The pressure side channels are disposed adjacent the pressure side wall, and the suction side channels are disposed adjacent the suction side wall.



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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation-in-part application of U.S. application Ser. No. 18/446,532 filed Aug. 9, 2023, which is hereby incorporated by reference in its entirety.

FIELD

[0002] The present disclosure relates generally to an airfoil for a turbomachine having a trailing edge cooling circuit. Particularly, the present disclosure is related to a trailing edge cooling circuit having separate pressure side and suction side cooling channels.

BACKGROUND

[0003] Turbomachines are utilized in a variety of industries and applications for energy transfer purposes. For example, a gas turbine engine generally includes a compressor section, a combustion section, a turbine section, and an exhaust section. The compressor section progressively increases the pressure of a working fluid entering the gas turbine engine and supplies this compressed working fluid to the combustion section. The compressed working fluid and a fuel (e.g., natural gas) mix within the combustion section and burn in a combustion chamber to generate high pressure and high temperature combustion gases. The combustion gases flow from the combustion section into the turbine section where they expand to produce work. For example, expansion of the combustion gases in the turbine section may rotate a rotor shaft connected, e.g., to a generator to produce electricity. The spent combustion gases then exit the turbine section via the exhaust section.

[0004] During operation of the turbomachine, various hot gas path components in the system are subjected to high temperature flows, which can stress the hot gas path components and shorten their useful life. Since higher temperature flows generally result in increased performance, efficiency, and power output of the turbomachine, the hot gas path components that are subjected to high temperature flows must be cooled to allow the gas turbine system to operate with flows at increased temperatures.

[0005] As the maximum local temperature of the hot gas path components approaches the melting temperature of the hot gas path components, forced air cooling becomes necessary. For this reason, airfoils of turbine rotor blades and stationary nozzles often require complex cooling schemes in which air, typically bleed air from the compressor section, is forced through internal cooling passages within the airfoil and then discharged through cooling holes at the airfoil surface to transfer heat from the hot gas path component.

[0006] The trailing edge region of the airfoil generally experiences higher thermal stresses than other regions of the airfoil. Traditional trailing edge regions often include a pin bank (or pin array), which includes a plurality of pins each extending directly between a pressure side wall and a suction side wall of the airfoil. However, issues exist with the use of pin banks. The pressure side wall often experiences higher temperatures than the suction side wall, and pin banks provide direct and discrete connections between the pressure side and the suction side, such that heat is conductively transferred from the pressure side to the suction side. This is disadvantageous because it places undesired thermal stresses on individual pins of the pin bank, particularly those pins in the vicinity of the airfoil platform. Additionally, the pins within the pin bank geometry are relatively weak structures when compared to the rest of the geometry of the airfoil. As such, an improved trailing edge cooling circuit that redistributes and reduces operational stresses in the trailing edge region without the use of a pin bank is desired.

BRIEF DESCRIPTION

[0007] Aspects and advantages of the airfoils and turbomachines in accordance with the present disclosure will be set forth in part in the following description, or may be obvious from the description, or may be learned through practice of the technology.

[0008] In accordance with one embodiment, an airfoil is provided. The airfoil includes a leading edge, a trailing edge, a pressure side wall extending between the leading edge and the trailing edge, and a suction side wall extending between the leading edge and the trailing edge. The pressure side wall and the suction side wall define a main body region and a trailing edge region. The main body region extends from the leading edge to the trailing edge region, and the trailing edge region extends from the main body region to the trailing edge. The main body region defines an upstream plenum extending along a span of the airfoil. The trailing edge region defines a plurality of pressure side channels and a plurality of suction side channels in fluid communication with the upstream plenum and arranged in a trussed structure. The plurality of pressure side channels is disposed adjacent the pressure side wall, and the plurality of suction side channels is disposed adjacent the suction side wall.

[0009] In accordance with another embodiment, a turbomachine is provided. The turbomachine includes a compressor section, a combustion section disposed downstream of the compressor section, and a turbine section disposed downstream of the combustion section. A rotor blade is disposed in one of the compressor section or the turbine section. The rotor blade includes a shank portion and an airfoil extending from the shank portion. The airfoil includes a leading edge, a trailing edge, a pressure side wall extending between the leading edge and the trailing edge, and a suction side wall extending between the leading edge and the trailing edge. The pressure side wall and the suction side wall define a main body region and a trailing edge region. The main body region extends from the leading edge to the trailing edge region, and the trailing edge region extends from the main body region to the trailing edge. The main body region defines an upstream plenum extending along a span of the airfoil. The trailing edge region defines a plurality of pressure side channels and a plurality of suction side channels in fluid communication with the upstream plenum and arranged in a trussed structure. The plurality of pressure side channels is disposed adjacent the pressure side wall, and the plurality of suction side channels is disposed adjacent the suction side wall.

[0010] These and other features, aspects and advantages of the present airfoils and turbomachines will become better understood with reference to the following description and appended claims. The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate embodiments of the technology and, together with the description, serve to explain the principles of the technology.

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## Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] A full and enabling disclosure of the present airfoils and turbomachines, including the best mode of making and using the present systems and methods, directed to one of ordinary skill in the art, is set forth in the specification, which makes reference to the appended figures, in which:

[0012] FIG. **1** is a schematic illustration of a turbomachine in accordance with embodiments of the present disclosure;

[0013] FIG. **2** illustrates a perspective, partial cut-away, view of a rotor blade in accordance with embodiments of the present disclosure;

[0014] FIG. **3** illustrates a cross-sectional perspective view of a portion of a trailing edge region of an airfoil, which includes a trailing edge cooling circuit, in accordance with embodiments of the present disclosure;

[0015] FIG. **4** illustrates a cross-sectional view, in a radial-transverse plane, of a trailing edge region of an airfoil having a trailing edge cooling circuit in accordance with embodiments of the present disclosure;
[0016] FIG. **5** illustrates a cross-sectional view, in a radial-transverse plane, of a trailing edge region of an airfoil having a trailing edge cooling circuit in accordance with embodiments of the present disclosure;
[0017] FIG. **6** illustrates a cross-sectional view of the trailing edge region of the airfoil of FIG. **4** from along the line **6-6** in accordance with embodiments of the present disclosure;
[0018] FIG. **7** illustrates a cross-sectional view of a trailing edge region of an airfoil in accordance with embodiments of the present disclosure;
[0019] FIG. **8** illustrates a cross-sectional view of a trailing edge region of an airfoil in accordance with embodiments of the present disclosure;
[0020] FIG. **9** illustrates a cross-sectional view of a trailing edge region of an airfoil in accordance with embodiments of the present disclosure;
[0021] FIG. **10** illustrates a cross-sectional view, in a radial-transverse plane, of a trailing edge region of an airfoil having a trailing edge cooling circuit in accordance with embodiments of the present disclosure;
[0022] FIG. **11** illustrates a cross-sectional view of the trailing edge region of the airfoil of FIG. **10** from along the line **11-11** in accordance with embodiments of the present disclosure;
[0023] FIG. **12** illustrates a perspective view of a rotor blade in accordance with embodiments of the present disclosure;
[0024] FIG. **13** illustrates a top view of a trailing edge portion of the airfoil of the rotor blade of FIG. **12**; and
[0025] FIG. **14** illustrates a perspective, cross-sectional view of a trailing edge portion of the rotor blade of FIG. **12**.

DETAILED DESCRIPTION

[0026] Reference now will be made in detail to embodiments of the present airfoils and turbomachines, one or more examples of which are illustrated in the drawings. Each example is provided by way of explanation, rather than limitation of, the technology. In fact, it will be apparent to those skilled in the art that modifications and variations can be made in the present technology without departing from the scope or spirit of the claimed technology. For instance, features illustrated or described as part of one embodiment can be used with another embodiment to yield a still further embodiment. Thus, it is intended that the present disclosure covers such modifications and variations as come within the scope of the appended claims and their equivalents.
[0027] The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any implementation described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other implementations. Additionally, unless specifically identified otherwise, all embodiments described herein should be considered exemplary.
[0028] The detailed description uses numerical and letter designations to refer to features in the drawings. Like or similar designations in the drawings and description have been used to refer to like or similar parts of the invention. As used herein, the terms “first”, “second”, and “third” may be used interchangeably to distinguish one component from another and are not intended to signify location or importance of the individual components.
[0029] The term “fluid” may be a gas or a liquid. The term “fluid communication” means that two or more areas defining a flow passage are joined to one another such that a fluid is capable of making the connection (i.e., flowing) between the areas specified.
[0030] As used herein, the terms “upstream” (or “forward”) and “downstream” (or “aft”) refer to the relative direction with respect to fluid flow in a fluid pathway. For example, “upstream” refers to the direction from which the fluid flows, and “downstream” refers to the direction to which the

fluid flows. However, the terms “upstream” and “downstream” as used herein may also refer to a flow of electricity. The term “radially” refers to the relative direction that is substantially perpendicular to an axial centerline of a particular component, the term “axially” refers to the relative direction that is substantially parallel and/or coaxially aligned to an axial centerline of a particular component, and the term “circumferentially” refers to the relative direction that extends around the axial centerline of a particular component.

[0031] Terms of approximation, such as “about,” “approximately,” “generally,” and “substantially,” are not to be limited to the precise value specified. In at least some instances, the approximating language may correspond to the precision of an instrument for measuring the value, or the precision of the methods or machines for constructing or manufacturing the components and/or systems. For example, the approximating language may refer to being within a 1, 2, 4, 5, 10, 15, or 20 percent margin in either individual values, range(s) of values and/or endpoints defining range(s) of values. When used in the context of an angle or direction, such terms include within ten degrees greater or less than the stated angle or direction. For example, “generally vertical” includes directions within ten degrees of vertical in any direction, e.g., clockwise or counter-clockwise.

[0032] The terms “coupled,” “fixed,” “attached to,” and the like refer to both direct coupling, fixing, or attaching, as well as indirect coupling, fixing, or attaching through one or more intermediate components or features, unless otherwise specified herein. The terms “directly coupled,” “directly fixed,” “directly attached to,” and the like mean that two components are joined in contact with one another and that no intermediate components or features are present.

[0033] As used herein, the terms “comprises,” “comprising,” “includes,” “including,” “has,” “having” or any other variation thereof, are intended to cover a non-exclusive inclusion. For example, a process, method, article, or apparatus that comprises a list of features is not necessarily limited only to those features but may include other features not expressly listed or inherent to such process, method, article, or apparatus. Further, unless expressly stated to the contrary, “and/or” refers to a condition satisfied by any one of the following: A is true (or present) and B is false (or not present), A is false (or not present) and B is true (or present), and both A and B are true (or present).

[0034] Here and throughout the specification and claims, where range limitations are combined and interchanged, such ranges are identified and include all the sub-ranges contained therein unless context or language indicates otherwise. For example, all ranges disclosed herein are inclusive of the endpoints, and the endpoints are independently combinable with each other.

[0035] Referring now to the drawings, FIG. **1** illustrates a schematic diagram of one embodiment of a turbomachine, which in the illustrated embodiment is a gas turbine engine **10**. Although an industrial or land-based gas turbine engine is shown and described herein, the present disclosure is not limited to an industrial or land-based gas turbine engine, unless otherwise specified in the claims. For example, the invention as described herein may be used in any type of turbomachine including, but not limited to, a steam turbine, an aircraft gas turbine, or a marine gas turbine.

[0036] As shown in FIG. **1**, the gas turbine engine **10** generally includes a compressor section **12**. The compressor section **12** includes a compressor **14**. The compressor section **12** includes an inlet **16** that is disposed at an upstream end of the gas turbine engine **10**. The gas turbine engine **10** further includes a combustion section **18** having one or more combustors **20** disposed downstream from the compressor section **12**. The gas turbine engine **10** further includes a turbine section **22** (i.e., an expansion turbine) that is downstream from the combustion section **18**. A shaft **24** extends generally axially through the gas turbine engine **10** and couples the compressor section **12** and the turbine section **22**.

[0037] The compressor section **12** may generally include a plurality of rotor disks **21** and a plurality of rotor blades **23** extending radially outwardly from and connected to each rotor disk **21**. Each rotor disk **21** in turn may be coupled to or form a forward portion of the shaft **24** that extends through the compressor section **12**. The rotor blades **23** of the compressor section **12** may include turbomachine airfoils that define an airfoil shape (e.g., having a leading edge, a trailing edge, and side walls extending between the leading edge and the trailing edge). Additionally, the compressor section **12** includes stator vanes **19** disposed between the rotor blades **23**. The stator vanes **19** may extend from and couple to a compressor casing **11**.

[0038] The turbine section **22** may generally include a plurality of rotor disks **27** and a plurality of rotor blades **28** extending radially outwardly from and being interconnected to each rotor disk **27**. Each rotor disk **27** in turn may be coupled to or form an aft portion of the shaft **24** that extends through the turbine section **22**. The turbine section **22** further includes an outer casing **32** that circumferentially surrounds the aft portion of the shaft **24** and the rotor blades **28**. The turbine section **22** may include stator vanes or stationary nozzles **26** extending radially inward from the outer casing **32**. The rotor blades **28** and stator vanes **26** may be arranged in alternating fashion in stages along an axial centerline **30** of gas turbine engine **10**. Both the rotor blades **28** and the stator vanes **26** may include turbomachine airfoils that define an airfoil shape (e.g., having a leading edge, a trailing edge, and side walls extending between the leading edge and the trailing edge).

[0039] In operation, ambient air or other working fluid is drawn into the inlet **16** of the compressor **14** and is progressively compressed to provide a compressed air **35** to the combustion section **18**. The compressed air **35** flows into the combustion section **18** and is mixed with fuel to form a combustible mixture. The combustible mixture is burned within a combustion chamber **25** of the combustor **20**, thereby generating combustion gases **41** that flow from the combustion chamber **25** into the turbine section **22**. Energy (kinetic and/or thermal) is transferred from the combustion gases **41** to the rotor blades **28**, causing the shaft **24** to rotate and produce mechanical work. The spent combustion gases **41** (also called “exhaust gases”) exit the turbine section **22** and flow through the exhaust diffuser **34** across a plurality of struts or main airfoils **43** that are disposed within the exhaust diffuser **34**.

[0040] The gas turbine engine **10** may define a cylindrical coordinate system having an axial direction A extending along the axial centerline **30**, a radial direction R perpendicular to the axial centerline **30**, and a circumferential direction C extending around the axial centerline **30**.

[0041] FIG. **2** provides a perspective, partial cut-away view of an exemplary rotor blade **50**. The rotor blade **50** may be the rotor blade **23** disposed in the compressor section **12** or the rotor blade **28** disposed in the turbine section **22**, which are described above with reference to FIG. **1**. As shown in FIG. **2**, the rotor blade **50** generally includes a shank portion **37** and an airfoil **40** that extends outwardly from the shank portion **37**. For example, the shank portion **37** may include a mounting portion **38** and a platform **42**, and the airfoil **40** may extend along the radial direction R from the platform **42**. The platform **42** generally serves as the radially inward boundary for the gases flowing through the gas turbine engine **10** (e.g., air flowing through the compressor section **12** or combustion gases **41** flowing through the hot gas path of the turbine section **22**, as shown in FIG. **1**). The platform **42** extends along the axial direction A from a leading face **84** to a trailing face **82**. As shown in FIG. **2**, the mounting portion **38** of the shank portion **37** may extend radially inwardly from the platform **42** and may include a root structure, such as a dovetail, configured to interconnect or secure the rotor blade **50** to a rotor disk **21**, **27** (FIG. **1**). In exemplary embodiments, the rotor blade **50** may be a turbine rotor blade (such as the rotor blade **28** described above with reference to FIG. **1**), which may benefit from the present cooling circuit(s).

[0042] The airfoil **40** includes a pressure side wall **44** and an opposing suction side wall **46**. The pressure side wall **44** and the suction side wall **46** extend substantially radially outwardly from the platform **42** in span from a root **48** of the airfoil **40**, which may be defined at an intersection between the airfoil **40** and the platform **42**, to a tip **51** of the airfoil **40**. The pressure side wall **44** is connected to the suction side wall **46** at a leading edge **52** of the airfoil **40** and a trailing edge **54** downstream of the leading edge **52**, and the airfoil **40** thus extends between the leading edge **52** and the trailing edge **54**. The pressure side wall **44** generally comprises an aerodynamic, concave external surface of the airfoil **40**. Similarly, the suction side wall **46** may generally define an aerodynamic, convex external surface of the airfoil **40**. The tip **51** is disposed radially opposite the root **48**. As such, the tip **51** may generally define the radially outermost portion of the rotor blade **50** and, thus, may be configured to be positioned adjacent to a stationary shroud or seal (not shown) of the gas turbine engine **10**. The tip **51** may include a tip cavity **66** or a tip shroud (not shown).

[0043] As shown in FIG. **2**, the rotor blade **50** may be at least partially hollow, e.g., the rotor blade **50** may include a cooling circuit **72** defined therein. The cooling circuit **72** may include a plurality of cooling passages **56** (shown partially in dashed lines in FIG. **2**), which may be circumscribed within the rotor blade **50** for routing a coolant **58** through the airfoil **40** between the pressure side wall **44** and the suction side wall **46**, thus providing convective cooling thereto. The cooling passages **56** may be at least partially defined by and between a plurality of ribs **74**. The ribs **74** extend partially through the cooling circuit **72** generally along the radial direction R, e.g., as illustrated in FIG. **2**. The ribs **74** may extend fully through the cooling circuit **72** between the pressure side wall **44** and the suction side wall **46**. The plurality of ribs **74** may thereby partition the cooling circuit **72** and at least partially form or define the cooling passages **56**. For example, each rib **74** may radially terminate near one of a root turn **76** or a tip turn **78**. The root turn **76** may be partially defined by a floor **77**, which defines the most radially inward boundary of the root turn **76**.

[0044] The coolant **58** may include a portion of the compressed air from the compressor section **12** (FIG. **1**) and/or steam or any other suitable gas or other fluid for cooling the airfoil **40**. One or more cooling passage inlets **60** are disposed along the rotor blade **50**. In some embodiments, one or more cooling passage inlets **60** are formed within, along or by the mounting portion **38**. The cooling passage inlets **60** are in fluid communication with at least one corresponding cooling passage **56**. A plurality of coolant outlets **64** may be in fluid communication with the tip cavity **66**. Each cooling passage **56** is in fluid communication with at least one of the coolant outlets **64**. In some embodiments, the tip cavity **66** may be at least partially surrounded by a pressure side tip rail **68** and a suction side tip rail **70**.

[0045] As may be seen in FIG. **2**, the cooling passages **56** extend within each of the shank portion **37** and the airfoil **40**. For example, the cooling passages **56** may extend between the shank portion **37** and the airfoil **40**, e.g., from the shank portion **37** to the airfoil **40**, such as from the one or more cooling passage inlets **60** in the shank portion **37** to the at least one coolant outlet **64** in the tip **51** of the airfoil **40**.

[0046] The airfoil **40** may define a camber axis **94** halfway between the pressure side wall **44** and the suction side wall **46**. The camber axis **94** may extend between (and intersect) the leading edge **52** and the trailing edge **54**. That is, the camber axis **94** may be an imaginary line that extends from the leading edge **52** to the trailing edge **54**, and the camber axis **94** may be the average curvature of the pressure side wall **44** and the suction side wall **46**. The camber axis **94** may be curved and/or contoured to correspond with the curve of the pressure side wall **44** and the suction side wall **46**. A transverse direction T may be defined orthogonally with respect to the camber axis **94**.

[0047] The cooling passages **56** may include a leading edge passage **88**, one or more intermediate passages **90**, and a trailing edge passage **92**. The leading edge passage **88** may be defined by the leading edge **52** and a rib **74**. The leading edge passage **88** may be the cooling passage **56** that is disposed closest to the leading edge **52**. Each of the intermediate passages **90** may be defined between two ribs **74**. The intermediate passages **90** may be disposed between the leading edge passage **88** and the trailing edge passage **92**. The trailing edge passage **92** may be defined by a rib **74** and a trailing edge cooling circuit **100**.

[0048] In many embodiments, the airfoil **40** may include a main body region **101** and a trailing edge region **102**. The main body region **101** and the trailing edge region **102** may extend over the entire span of the airfoil **40** (e.g., radially between the root **48** and the tip **51**). The main body region **101** may extend along the camber axis **94** between the leading edge **52** and the trailing edge region **102**, and the trailing edge region **102** may extend along the camber axis **94** from the main body region **101** to the trailing edge **54**. Additionally, the trailing edge region **102** may extend along the camber axis **94** from the trailing edge **54** to between about 10% and about 40% of an entire length of the camber axis **94**, or such as between about 10% and about 35% of the entire length of the camber axis **94**, or such as between about 10% and about 30% of the entire length of the camber axis **94**, or such as between about 10% and about 25% of the entire length of the camber axis **94**, or such as between about 10% and about 20% of the entire length of the camber axis **94**. As should be appreciated, the trailing edge region **102** may generally be exposed to higher thermal stresses during operation of the gas turbine engine **10**. The trailing edge cooling circuit **100** may be disposed in the trailing edge region **102**, in order to provide convective cooling to the trailing edge region **102**, thereby increasing the hardware life of the airfoil **40**.

[0049] Referring now to FIG. **3**, a cross-sectional perspective view of a trailing edge region **102** of the airfoil **40** having a trailing edge cooling circuit **100** is illustrated in accordance with embodiments of the present disclosure. As shown, the trailing edge cooling circuit **100** includes a plurality of pressure side channels **104** and a plurality of suction side channels **106**. The pressure side channels **104** and the suction side channels **106** may be radially spaced apart from one another. Moreover, the plurality of pressure side channels **104** and the plurality of suction side channels **106** may be fluidly isolated. For example, a plurality of shared walls **108** may radially separate the plurality of pressure side channels **104** and the plurality of suction side channels **106**. Particularly, the pressure side channel **104** and the suction side channel **106** may each be partially defined by a shared wall **108** of the plurality of shared walls **108**. The shared walls **108** create a trussed structure that distributes stress across the shared walls **108** and the trailing edge region **102**.

[0050] For example, each shared wall **108** may define a pressure side channel **104** of the plurality of pressure side channels **104** and a neighboring suction side channel **106** of the plurality of suction side channels **106**. Each shared wall **108** of the plurality of shared walls **108** may extend generally oblique to the radial direction R between an interior of the suction side wall **46** and an interior of the pressure side wall **44**. In some embodiments, each shared wall **108** may slope radially outwardly as the shared wall **108** extends between one of the pressure side wall **44** or the suction side wall **46** to the other of the pressure side wall **44** or the suction side wall **46**.

[0051] Each pressure side channel **104** of the plurality of pressure side channels **104** may extend from a pressure side inlet **118** to a pressure side outlet **120**. Likewise, each suction side channel **106** of the plurality of suction side channels **106** may extend from a suction side inlet **122** to a suction side outlet **124**. As shown in FIG. **3**, the pressure side channel **104** may converge in cross-sectional area as the pressure side channel **104** extends between the pressure side inlet **118** and the pressure side outlet **120**. Additionally, the suction side channel **106** may converge in cross-sectional area between the suction side inlet **122** and the suction side outlet **124**. Particularly, both a radial length and transverse length (e.g., perpendicular to the camber axis **94**) of the channels **104**, **106** may converge (or decrease) as the channels **104**, **106** extend towards the trailing edge **54**.

[0052] In at least one example embodiment, because the plurality of pressure side channels **104** and the plurality of suction side channels **106** are fluidly isolated, a geometry of the pressure side channels **104** and a geometry of the suction side channels **106** may be customized independently of the other. For example, a size, shape, radial length, transverse length, and/or cross-sectional area of one or more of the plurality of pressure side channels **104** may be modified or different than another one or more of the plurality of pressure side channels **104** and/or one of more of the plurality of suction side channels **106**. Similarly, a size, shape, radial length, transverse length, and/or cross-sectional area of one or more of the plurality of suction side channels **106** may be modified or different than another one or more of the plurality of suction side channels **106** and/or one of more of the plurality of pressure side channels **104**. Regardless of differences in geometry, in one embodiment, the plurality of pressure side channels **104** and the plurality of suction side channels **106** include walls that extend from one of the pressure side wall **44** or the suction side wall **46** to the other of the pressure side wall **44** or the suction side wall **46**, thus creating a trussed structure. As discussed previously, in some embodiments, the plurality of pressure side channels

**104** and the plurality of suction side channels **106** are partially defined by shared walls **108** that are common to one of the pressure side channels **104** and an adjacent one of the suction side channels **106**.
[0053] Referring now to FIG. **4**, a cross-sectional view of the trailing edge region **102** of the airfoil **40** having the trailing edge cooling circuit **100** in a radial-transverse plane is illustrated in accordance with embodiments of the present disclosure. As shown, the plurality of shared walls **108** may include a plurality of first shared walls **110** and a plurality of second shared walls **112**. In forming a trussed structure, the plurality of first shared walls **110** may each extend obliquely to the radial direction R from the pressure side wall **44** to the suction side wall **46**, and the plurality of second shared walls **112** may extend obliquely to the radial direction R from the suction side wall **46** to the pressure side wall **44**. Each of the first shared walls **110** may intersect (at a first intersection point **114**) one of the second shared walls **112** at the pressure side wall **44** and may intersect (at a second intersection point **116**) a different one of the second shared walls **112** at the suction side wall **46**. Stated otherwise, each of the second shared walls **112** may intersect (at the first intersection point **114**) one of the first shared walls **110** at the pressure side wall **44** and may intersect (at the second intersection point **116**) a different one of the first shared walls **110** at the suction side wall **46**. The extension of the walls **112**, **114** between the pressure side and suction side walls **44**, **46** distributes stress and reduces discrete stress locations that may be experienced with conventional pin banks.
[0054] The pressure side channel **104** may be collectively defined by a first shared wall **110** of the plurality of first shared walls **110**, a second shared wall **112** of the plurality of second shared walls **112**, and an interior of the pressure side wall **44**. Similarly, the suction side channel **106** may be collectively defined by a first shared wall **110** of the plurality of first shared walls **110**, a second shared wall **112** of the plurality of second shared walls **112**, and an interior of the suction side wall **46**.
[0055] In some embodiments, as shown in FIG. **4**, the pressure side channel **104** and the suction side channel **106** may define a triangle-shaped area **109** (e.g., cross-sectional area). Particularly, the pressure side channel **104** and the suction side channel **106** may define a triangle-shaped area **109** in the radial-transverse plane. Referring back to FIG. **3** briefly, the pressure side channel **104** may converge in cross-sectional area between the pressure side inlet **118** and the pressure side outlet **120**. Similarly, the suction side channel **106** may converge in cross-sectional area between the suction side inlet **122** and the suction side outlet **124**. For example, the triangle-shaped area **109** may converge (or continually decrease) as the pressure side channel **104** and the suction side channel **106** extend from the respective inlet **118**, **122** to the respective outlet **120**, **124**.
[0056] Referring now to FIG. **5**, a cross-sectional view of the trailing edge region **102** of the airfoil **40** having the trailing edge cooling circuit **100** in a radial-transverse plane is illustrated in accordance with another embodiment of the present disclosure. As shown, the trailing edge cooling circuit **100** may include a plurality of pressure side channels **206** and a plurality of suction side channels **208**, which are defined by conduits (not separately numbered). Particularly, the pressure side channels **206** and the suction side channels **208** may be arranged in radial rows **212**, **214**. For example, the plurality of pressure side channels **206** may be disposed in a pressure side radial row **212**, and the plurality of suction side channels **208** may be disposed in a suction side row **214**. Particularly, the plurality of pressure side channels **206** may each be centered along a first radial line **216** proximate to the pressure side wall **44** (e.g., closer to the pressure side wall **44** than the suction side wall **46**). Similarly, the plurality of suction side channels **208** may each be centered along a second radial line **218** proximate to the suction side wall **46** (e.g., closer to the suction side wall **46** than the pressure side wall **44**).
[0057] Additionally, as shown in FIG. **5**, each of the pressure side channels **206** and the suction side channels **208** may define an oval-shaped cross-sectional area **209**. Particularly, the oval-shaped area **209** may be defined in the radial-transverse plane, and the oval-shaped area **209** may include a

major axis **220** and a minor axis **222**. The major axis **220** may be the longest dimension of the oval-shaped area **209**, and the minor axis **222** may be the shortest dimension of the oval-shaped area **209**. The major axis **220** may be parallel to the radial direction R, and the minor axis **222** may be parallel to the transverse direction T.

[0058] In many embodiments, as shown in FIG. **5**, the plurality of suction side channels **208** may be radially staggered relative to the plurality of pressure side channels **206**, such that each suction side channel **208** radially overlaps with at least one (or two) pressure side channels **206** of the plurality of pressure side channels **206**. For example, each suction side channel **208** may be disposed radially between two pressure side channels **206** of the plurality of pressure side channels **206**. Similarly, each pressure side channel **206** may be disposed radially between two suction side channels **208** of the plurality of suction side channels **208**. The channels **206**, **208** may be defined through a solid material that connects the pressure side wall **44** and the suction side wall **46** (as shown), or each channel **206**, **208** may be defined by a channel wall that circumscribes the respective channel **206**, **208** and that is coupled to a channel wall of a transversely adjacent channel **208**, **206**, thereby creating a stress-mitigating lattice.

[0059] Referring now to FIG. **6**, a cross-sectional view of the trailing edge region **102** of the airfoil **40**, which includes a trailing edge cooling circuit **100**, from along the line **6-6** shown in FIG. **4** is illustrated in accordance with embodiments of the present disclosure. The trailing edge region **102** of the airfoil **40** may include a portion of the suction side wall **46** and a portion of the pressure side wall **44**, and the camber axis **94** extends between the suction side wall **46** and the pressure side wall **44** through the trailing edge region **102** to the trailing edge **54**. An axial direction A.sub.c may extend along the camber axis **94**, and the transverse direction T may be perpendicular to the axial direction A.sub.c. As shown in FIG. **6**, the trailing edge cooling circuit **100** may include a pressure side channel **104** and a suction side channel **106** (shown in phantom).

[0060] The pressure side channel **104** may be disposed at least partially on a first side (e.g., a first transverse side) of the camber axis **94**, and the suction side channel **106** may be disposed at least partially on a second side (e.g., a second transverse side) of the camber axis **94**. The first side may be transversely between the camber axis **94** and the pressure side wall **44**, and the second side may be transversely between the camber axis **94** and the suction side wall **46**. Particularly, a majority (e.g., more than about 50%, or such as more than about 65%, or such as more than 75%, or such as more than 90%) of the pressure side channel **104** may be disposed on the first side of the camber axis **94**. Similarly, a majority (e.g., more than about 50%, or such as more than about 65%, or such as more than 75%, or such as more than 90%) of the suction side channel **106** may be disposed on the second side of the camber axis **94**.

[0061] In exemplary embodiments, as shown in FIG. **6**, the pressure side channel **104** may extend between a pressure side inlet **118** and a pressure side outlet **120** and along a pressure side centerline **126** on the first side of the camber axis **94**. Similarly, the suction side channel **106** extends between a suction side inlet **122** and a suction side outlet **124** and along a suction side centerline **128** on the second side of the camber axis **94**. In such embodiments, the pressure side centerline **126** may converge towards the camber axis **94** between the pressure side inlet **118** and the pressure side outlet **120**, and the suction side centerline **128** may converge towards the camber axis **94** between the suction side inlet **122** and the suction side outlet **124**. Thus, the pressure side channel **104** and the suction side channel **106** may be said to converge towards the trailing edge **54** as the pressure side channel **104** and the suction side channel **106** extend towards the trailing edge **54**.

[0062] In many embodiments, the pressure side channel **104** and the suction side channel **106** may each extend to, and be in direct fluid communication with, a common plenum, such as a downstream plenum **130**. For example, the pressure side channel **104** may extend from the pressure side inlet **118** to the pressure side outlet **120** at the downstream plenum **130**. Similarly, the suction side channel **106** may extend from the suction side inlet **122** to the suction side outlet **124** at the downstream plenum **130**. The downstream plenum **130** may be centered on the camber axis **94** (such that a centerline of the downstream plenum **130** is aligned with the camber axis **94**). In exemplary embodiments, as shown in FIGS. **3** and **6**, the pressure side channel **104** and the suction side channel **106** may be disposed with respective centerlines **126**, **126** in different radial (span-wise) planes and thus may converge on (intersect with) the downstream plenum **130** via outlets **120**, **124** located in different radial planes.

[0063] In exemplary embodiments, one or more exit channels **132** may extend from the downstream plenum **130** to corresponding outlet(s) **134** at the trailing edge **54**. The one or more exit channels **132** may extend directly along the camber axis **94** from the downstream plenum **130** to the trailing edge **54**, in order to exhaust coolant from the trailing edge cooling circuit **100** (and the entire cooling circuit **72**).

[0064] Referring now to FIG. **7**, a cross-sectional view of the trailing edge region **102** of the airfoil **40** is illustrated in accordance with another embodiment of the present disclosure. As shown, the pressure side channel **104** and the suction side channel **106** may be defined at least partially in the same axial-transverse plane. In such embodiments, a dividing wall **136** may separate, and partially define, the pressure side channel **104** and the suction side channel **106**. The dividing wall **136** may extend generally axially and be disposed transversely between the suction channel **106** and the pressure side channel **104**. Additionally, the dividing wall **136** may extend at least partially along the camber axis **94**. Other walls defining the channels **104**, **106** (e.g., walls spanning between the pressure side wall **44** and the suction side wall **46** in various radial planes) are not illustrated.

[0065] In various embodiments, the pressure side channel **104** may terminate forward (or upstream) of the trailing edge **54**, such that the suction side channel **106** extends axially beyond the pressure side channel **104**. For example, the suction side channel **106** may extend from the suction side inlet **122** to an exit channel **132** at the trailing edge **54**, and the pressure side channel **104** may extend from a pressure side inlet **118** to a pressure side exit channel **138**. The pressure side exit channel **138** may extend from the pressure side channel **104** to an outlet on the pressure side wall **44**. The pressure side exit channel **138** may be generally oblique to the camber axis **94** (and the pressure side wall **44**) in the axial-transverse plane, such that the pressure side exit channel **138** advantageously exhausts coolant at an angle for film cooling the pressure side wall **44** upstream of the trailing edge **54**.

[0066] Referring now to FIGS. **8** and **9**, a cross-sectional view of the trailing edge region **102** of the airfoil **40** is illustrated in accordance with embodiments of the present disclosure. As shown, the dividing wall **136** may include an axial portion **142** and a transverse portion **144**. The pressure side channel **104** may terminate forward (or upstream) of the trailing edge **54** at the transverse portion **144** of the dividing wall **136**, such that the suction side channel **106** extends axially beyond the pressure side channel **104**. In such embodiments, the suction side channel **106** may include a first portion **146** and a second portion **148**. The first portion **146** may extend from the suction side inlet **122** to the transverse portion **144** of the dividing wall **136**, and the second portion **148** may extend from the transverse portion **144** of the dividing wall **136** to an exit channel **132** and/or **152**. The first portion **146** may have a smaller transverse width than the second portion **148**. The first portion **146** may be defined (transversely) between the suction side wall **46** and the dividing wall **136**, and the second portion **148** may be defined (transversely) between the pressure side wall **44** and the suction side wall **46**.

[0067] As shown in FIGS. **8** and **9**, both the pressure side channel **104** and the suction side channel **106** may include separate pressure side exit channels. For example, the pressure side channel **104** may extend to a first pressure side exit channel **150**, and the suction side channel **106** may include a second pressure side exit channel **152** disposed downstream of the first pressure side exit channel **150**. The first pressure side exit channel **150** may extend generally obliquely (relative to the camber axis **94** and the pressure side wall **44**) from a terminal end of the pressure side channel **104** to an outlet on the pressure side wall **44**.

[0068] In some embodiments, as shown in FIG. **8**, the second pressure side exit channel **152** may extend generally obliquely (relative to the camber axis **94** and the pressure side wall **44**) from a terminal end of the suction side channel **106** to an outlet on the pressure side wall **44**. In such embodiments, the second pressure side exit channel **152** may be the only exit channel to the suction side channel **106** (e.g., the trailing edge cooling circuit **100** may not include an exit channel at the trailing edge **54**). In other embodiments, as shown in FIG. **9**, the suction side channel **106** may further include a trailing edge exit channel **132** downstream of the second pressure side exit channel **152** with respect to the flow of coolant (from left to right) through the trailing edge cooling circuit **100**.

[0069] Referring now to FIG. **10**, a cross-sectional view of the trailing edge region **102** of the airfoil **40** having the trailing edge cooling circuit **100** in a radial-transverse plane is illustrated in accordance with another embodiment of the present disclosure. As shown, the trailing edge cooling circuit **100** may include a plurality of pressure side channels **306** and a plurality of suction side channels **308**. Particularly, the pressure side channels **306** and the suction side channels **308** may be arranged in radial rows **312**, **314**. For example, the plurality of pressure side channels **306** may be disposed in a pressure side radial row **312**, and the plurality of suction side channels **308** may be disposed in a suction side radial row **314**. Particularly, the plurality of pressure side channels **306** may each be centered along a first radial line **316** proximate to the pressure side wall **44** (e.g., closer to the pressure side wall **44** than the suction side wall **46**). Similarly, the plurality of suction side channels **308** may each be centered along a second radial line **318** proximate to the suction side wall **46** (e.g., closer to the suction side wall **46** than the pressure side wall **44**).

[0070] Additionally, as shown in FIG. **10**, each of the pressure side channels **306** and the suction side channels **308** may define an oval-shaped area **309**. Particularly, the oval-shaped area **309** may be defined in the radial-transverse plane, and the oval-shaped area **309** may include a major axis **320** and a minor axis **322**. The major axis **320** may be the longest dimension of the oval-shaped area **309**, and the minor axis **322** may be the shortest dimension of the oval-shaped area **309**. The major axis **320** may be parallel to the radial direction R, and the minor axis **322** may be parallel to the transverse direction T.

[0071] In many embodiments, as shown in FIG. **10**, the plurality of suction side channels **308** may be radially staggered relative to the plurality of pressure side channels **306**, such that each suction side channel **308** radially overlaps with at least one (or two) pressure side channels **306** of the plurality of pressure side channels **306**. Additionally, as shown in FIG. **10**, an area (such as a cross-sectional area in the radial-transverse plane) of one of the pressure side channel **306** or the suction side channel **308** may be larger than an area (such as a cross-sectional area in the radial-transverse plane) of the other of the pressure side channel **306** or the suction side channel **308**. For example, the oval-shaped area **309** of one of the pressure side channels **306** or the suction side channels **308** may be larger than the oval-shaped area **309** of the other of the pressure side channels **306** or the suction side channels **308**. In the illustrated exemplary embodiment, the major axis **320** and the minor axis **322** of the oval-shaped area **309** of the suction side channels **308** may be longer than the major axis **320** and the minor axis **322** of the oval-shaped area **309** of the pressure side channels **306**. In this way, the flow of cooling air to the pressure side channels **306** and/or the suction side channels **308** may be metered based on cooling needs by adjusting the size of the channels **306**, **308**.

[0072] In various embodiments, the airfoil **40** may define dead space **324** that is disposed between the pressure side channels **306** and the suction side channels **308**. In some embodiments, the dead space **324** may be a cavity that extends radially the entire span of the airfoil **40** (e.g., from the root **48** to the tip **51**). In other embodiments, the dead space **324** may be a cavity that extends only a portion of the span of the airfoil **40**. Particularly, a first dividing wall **326** may extend radially through the airfoil **40** and partially define the dead space **324** and one or more of the pressure side channels **306**. A second dividing wall **328** may extend radially through the airfoil **40** and partially define the dead space **324** and one or more of the suction side channels **308**. That is, the dividing walls **326**, **328** may be spaced apart from one another in the transverse direction T, such that the dead space **324** is defined between the first dividing wall **326** and the second dividing wall **328**.

[0073] Referring now to FIG. **11**, a cross-sectional view of the trailing edge region **102** of the airfoil **40**, which includes a trailing edge cooling circuit **100**, from along the line **11-11** shown in FIG. **10** is illustrated in accordance with embodiments of the present disclosure. As shown, the pressure side channel **306** may be defined by the first dividing wall **326** and by an interior of the pressure side wall **44**. The suction side channel **308** may be defined by the second dividing wall **328** and by an interior of the suction side wall **46**. The channels **308** may each extend to a common plenum **330**, and an exit channel **332** may extend from the common plenum **330** to the trailing edge **54**. The dead space **324** may include an open end **334** and a closed end **336**, and the closed end **336** may partially define the common plenum **330**. In this way, air flowing into the dead space **324** does not immediately exit the airfoil **40** (e.g., from the dead space **324**). Rather, all air entering the dead space **324** must eventually exit the dead space **324** and enter one of the channels **306**, **308**.

[0074] As used herein, “in cooling proximity” refers to a relationship between a cooling channel (such as the suction side channel **106** or the pressure side channel **104**) and a respective surface along which a majority of the cooling channel is located (e.g., the suction side wall **46** and the pressure side wall **44**), such that the temperature of the respective surface is decreased by the fluid flowing through the cooling channel. For example, the pressure side channels **104** may be in cooling proximity with the pressure side wall **44** but not the suction side wall **46**. Similarly, the suction side channels **106** may be in cooling proximity with the suction side wall **46** but not the pressure side wall **44**.

[0075] FIG. **12** illustrates a perspective view of a rotor blade **1250** in accordance with embodiments of the present disclosure. The rotor blade **1250** may be similar or analogous to the rotor blade **50** discussed above with respect to FIGS. **2-11**. For example, the rotor blade **1250** may be incorporated into the gas turbine engine **10** in place of the rotor blade **23** disposed in the compressor section **12** or the rotor blade **28** disposed in the turbine section **22**, which are described above with reference to FIG. **1**.

[0076] As shown in FIG. **12**, the rotor blade **1250** generally includes a shank portion **1237** and an airfoil **1240** that extends outwardly from the shank portion **1237**. For example, the shank portion **1237** may include a mounting portion **1238** and a platform **1242**, and the airfoil **1240** may extend along the radial direction R (FIG. **2**) from the platform **1242**. The platform **1242** generally serves as the radially inward boundary for the gases flowing through the gas turbine engine **10** (e.g., air flowing through the compressor section **12** or combustion gases **41** flowing through the hot gas path of the turbine section **22**, as shown in FIG. **1**). The platform **1242** extends along the axial direction A (FIG. **2**) from a leading face **1284** to a trailing face **1282**. As shown in FIG. **12**, the mounting portion **1238** of the shank portion **1237** may extend radially inwardly from the platform **1242** and may include a root structure configured to interconnect or secure the rotor blade **1250** to a rotor disk **21**, **27** (FIG. **1**). The root structure may be a dovetail in some example embodiments. Additionally, the rotor blade **1250** may be a turbine rotor blade (such as the rotor blade **28** described above with reference to FIG. **1**) in some example embodiment, which may benefit from the present cooling circuit(s).

[0077] The airfoil **1240** includes a pressure side wall **1244** and a suction side wall **1246** opposite the pressure side wall **1244**. The pressure side wall **1244** and the suction side wall **1246** extend substantially radially outwardly from the platform **1242** and span from a root **1248** of the airfoil **1240**, which may be defined at an intersection between the airfoil **40** and the platform **42**, to a tip **1251** of the airfoil **1240**. The pressure side wall **1244** is connected to the suction side wall **1246** at a leading edge **1252** of the airfoil **1240** and a trailing edge **1254** downstream of the leading edge **1252**, and the airfoil **1240** thus extends between the leading edge **1252** and the trailing edge **1254**. The pressure side wall **1244** generally comprises an aerodynamic, concave external surface of the airfoil **1240**. Similarly, the suction side wall **46** may generally define an aerodynamic, convex external surface of the airfoil **1240**. The tip **1251** is disposed radially opposite the root **1248**. As such, the tip **1251** may generally define the radially outermost portion of the rotor blade **1250** and, thus, may be configured to be positioned adjacent to a stationary shroud or seal (not shown) of the gas turbine engine **10**. The tip **1251** may include a tip cavity, such as the tip cavity **66** (shown in FIG. **2**) or a tip shroud (not shown).

[0078] The airfoil **1240** may define a camber axis **1294** halfway between the pressure side wall **1244** and the suction side wall **1246**. The camber axis **1294** may extend between (and intersect) the leading edge **1252** and the trailing edge **1254**. That is, the camber axis **1294** may be an imaginary line that extends from the leading edge **1252** to the trailing edge **1254**, and the camber axis **1294** may be the average curvature of the pressure side wall **1244** and the suction side wall **1246**. The camber axis **1294** may be curved and/or contoured to correspond with the curve of the pressure side wall **1244** and the suction side wall **1246**.

[0079] In at least one example embodiment, the airfoil **1240** includes a main body region **1201** and a trailing edge region **1202**. The main body region **1201** and the trailing edge region **1202** may extend the entire span of the airfoil **1240** (e.g., radially between the root **1248** and the tip **1251**). The main body region **1201** may extend along the camber axis **1294** between the leading edge **1252** and the trailing edge region **1202**, and the trailing edge region **1202** may extend along the camber axis **1294** from the main body region **1201** to the trailing edge **1254**. Additionally, the trailing edge region **1202** may extend along the camber axis **1294** from the trailing edge **1254** to between about 10% and about 40% of an entire length of the camber axis **1294**, or such as between about 10% and about 35% of the entire length of the camber axis **1294**, or such as between about 10% and about 30% of the entire length of the camber axis **1294**, or such as between about 10% and about 25% of the entire length of the camber axis **1294**, or such as between about 10% and about 20% of the entire length of the camber axis **94**. As should be appreciated, the trailing edge region **1202** may generally be exposed to higher thermal stresses during operation of the gas turbine engine **10**. Accordingly, the trailing edge cooling circuit **100** may be disposed in the trailing edge region **1202**, in order to provide convective cooling to the trailing edge region **1202**, thereby increasing the hardware life of the airfoil **1240**.

[0080] FIG. **13** illustrates a top view of a portion of the airfoil **1240** of the rotor blade **1250** of FIG. **12**. FIG. **14** illustrates a perspective, cross-sectional view of a portion of the rotor blade **1250** of FIG. **12**.

[0081] As shown in FIGS. **12-14**, the main body region **1201** defines an upstream plenum **1205** that extends along a span (i.e., radial direction) of the airfoil **1240**. The upstream plenum **1205** is in fluid communication with the trailing edge region **1202**. For example, the upstream plenum **1205** may be configured to receive the coolant **58** and deliver the coolant **58** to the trailing edge cooling circuit **100** of the trailing edge region **1202**. The coolant **58** may include a portion of the compressed air from the compressor section **12** (FIG. **1**) and/or steam or any other suitable gas or other fluid for cooling the airfoil **40**.

[0082] With reference to FIG. **14**, the cooling circuit **100** disposed in the trailing edge region **1202** of the airfoil **1240** is similar or analogous to the cooling circuit **100** disposed in the trailing edge region **102** of the airfoil **40** discussed above with respect to FIGS. **3-11**. For example, the cooling circuit **100** includes the plurality of pressure side channels **104** and the plurality of suction side channels **106**. The plurality of pressure side channels **104** and the plurality of suction side channels **106** are in fluid communication with the upstream plenum **1205**, such that the coolant flow into the channels **104**, **106** is directed in a predominantly axial direction toward the trailing edge **1254** rather than being turned (e.g., around a rib **74**) and imparted with a partial radial directional component before entering the channels **104**, **106**.

[0083] As shown in FIG. **14**, the upstream plenum **1205** and the cooling circuit **100** are disposed in the airfoil **1240**. In some example embodiments, the upstream plenum **1205** and/or the cooling circuit **100** may at least partially extend into the shank portion **1237** of the rotor blade **1250**.

[0084] The trailing edge cooling circuit **100** described herein advantageously reduces peak thermal stresses experienced in the trailing edge region **102** by spreading stresses out over larger, connected structures (e.g., trussed structures) as opposed to concentrating stresses on single, individual pins of a pin bank. The shared walls and/or staggered cooling channels structurally share loading while also allowing heat to spread more gradually through the trailing edge region **102**, which reduces peak stresses. Additionally, the pressure side cooling channels **104** and the suction side cooling channels **106** may be shaped to have a large surface area adjacent the pressure side **44** and the suction side **46**, respectively, thereby promoting cooling. Accordingly, the design of the pressure side cooling channels **104** and the suction side cooling channels **106** may be customized to meet a desired need, such as to meet thermal demands of the airfoil **40**, **1240**, which may further reduce thermal stresses.
[0085] This written description uses examples to disclose the invention, including the best mode, and also to enable any person skilled in the art to practice the invention, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the invention is defined by the claims and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they include structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal language of the claims.
[0086] Further aspects of the invention are provided by the subject matter of the following clauses:
[0087] An airfoil comprising: a leading edge; a trailing edge; a pressure side wall and a suction side wall each extending between the leading edge and the trailing edge, wherein a camber axis is defined halfway between the pressure side wall and the suction side wall; and a cooling circuit defined in the airfoil, the cooling circuit having a trailing edge cooling circuit comprising a pressure side channel and a suction side channel, the pressure side channel disposed in cooling proximity to the pressure side wall and the suction side channel disposed in cooling proximity to the suction side wall.
[0088] The airfoil as in any preceding clause, wherein the pressure side channel is disposed at least partially on a first side of the camber axis and the suction side channel is disposed at least partially on a second side of the camber axis, and wherein the pressure side channel and the suction side channel each converge towards the trailing edge as the pressure side channel and the suction side channel extend towards the trailing edge.
[0089] The airfoil as in any preceding clause, wherein the pressure side channel and the suction side channel are partially defined by a shared wall disposed between the pressure side wall and the suction side wall.
[0090] The airfoil as in any preceding clause, wherein the pressure side channel extends between a pressure side inlet and a pressure side outlet and along a pressure side centerline on a first side of the camber axis, and wherein the suction side channel extends between a suction side inlet and a suction side outlet and along a suction side centerline on a second side of the camber axis.
[0091] The airfoil as in any preceding clause, wherein the pressure side centerline converges towards the camber axis between the pressure side inlet and the pressure side outlet, and wherein the suction side centerline converges towards the camber axis between the suction side inlet and the suction side outlet.
[0092] The airfoil as in any preceding clause, wherein the pressure side channel and the suction side channel are radially spaced apart from one another.
[0093] The airfoil as in any preceding clause, wherein the pressure side channel converges in cross-sectional area between the pressure side inlet and the pressure side outlet, wherein the suction side channel converges in cross-sectional area between the suction side inlet and the suction side outlet.
[0094] The airfoil as in any preceding clause, wherein the pressure side channel and the suction side channel each extend to a common plenum.
[0095] The airfoil as in any preceding clause, wherein one or more exit channels extend from the

common plenum to the trailing edge.
[0096] The airfoil as in any preceding clause, wherein the pressure side channel extends to a pressure side exit channel.
[0097] The airfoil as in any preceding clause, wherein the pressure side exit channel is a first pressure side exit channel, and wherein the suction side channel includes a second pressure side exit channel disposed downstream of the first pressure side exit channel.
[0098] The airfoil as in any preceding clause, wherein the pressure side channel and the suction side channel define one of a triangle-shaped or oval-shaped cross-sectional area.
[0099] The airfoil as in any preceding clause, wherein the pressure side channel is one of a plurality of pressure side channels, wherein the suction side channel is one of a plurality of suction side channels, and wherein the plurality of suction side channels is radially staggered relative to the plurality of pressure side channels such that each suction side channel radially overlaps with at least one pressure side channel of the plurality of pressure side channels.
[0100] The airfoil as in any preceding clause, wherein the airfoil further defines a dead space disposed between the pressure side channel and the suction side channel.
[0101] The airfoil as in any preceding clause, wherein an area of one of the pressure side channel or the suction side channel is larger than an area of the other of the pressure side channel or the suction side channel.
[0102] A turbomachine comprising: a compressor section; a combustion section disposed downstream of the compressor section; and a turbine section disposed downstream of the combustion section, wherein a rotor blade is disposed in one of the combustion section or the turbine section, the rotor blade having an airfoil that comprises: a leading edge; a trailing edge; a pressure side wall and a suction side wall each extending between the leading edge and the trailing edge, wherein a camber axis is defined halfway between the pressure side wall and the suction side wall; and a cooling circuit defined in the airfoil, the cooling circuit having a trailing edge cooling circuit comprising a pressure side channel and a suction side channel, the pressure side channel disposed in cooling proximity to the pressure side wall and the suction side channel disposed in cooling proximity to the suction side wall.
[0103] The turbomachine as in any preceding clause, wherein the pressure side channel is disposed at least partially on a first side of the camber axis and the suction side channel is disposed at least partially on a second side of the camber axis, and wherein the pressure side channel and the suction side channel each converge towards the trailing edge as the pressure side channel and the suction side channel extend towards the trailing edge.
[0104] The turbomachine as in any preceding clause, wherein the pressure side channel and the suction side channel are partially defined by a shared wall disposed between the pressure side wall and the suction side wall.
[0105] The turbomachine as in any preceding clause, wherein the pressure side channel extends between a pressure side inlet and a pressure side outlet and along a pressure side centerline on a first side of the camber axis, and wherein the suction side channel extends between a suction side inlet and a suction side outlet and along a suction side centerline on a second side of the camber axis.
[0106] The turbomachine as in any preceding clause, wherein the pressure side centerline converges towards the camber axis between the pressure side inlet and the pressure side outlet, and wherein the suction side centerline converges towards the camber axis between the suction side inlet and the suction side outlet.
[0107] An airfoil comprising: a leading edge; a trailing edge; a pressure side wall extending between the leading edge and the trailing edge; and a suction side wall extending between the leading edge and the trailing edge; wherein the pressure side wall and the suction side wall define a main body region and a trailing edge region, the main body region extending from the leading edge to the trailing edge region and the trailing edge region extending from the main body region to the

trailing edge; wherein the main body region defines an upstream plenum extending along a span of the airfoil; and wherein the trailing edge region defines a plurality of pressure side channels and a plurality of suction side channels in fluid communication with the upstream plenum and arranged in a trussed structure; and wherein the plurality of pressure side channels is disposed adjacent to the pressure side wall, and the plurality of suction side channels is disposed adjacent to the suction side wall.

[0108] The airfoil as in any preceding clause, wherein: a camber axis is defined halfway between the pressure side wall and the suction side wall; the plurality of pressure side channels is disposed at least partially on a first side of the camber axis and the plurality of suction side channels is disposed at least partially on a second side of the camber axis; and the plurality of pressure side channels and the plurality of suction side channels each converge towards the trailing edge as the plurality of pressure side channels and the plurality of suction side channels extend towards the trailing edge.

[0109] The airfoil as in any preceding clause, wherein each of the plurality of pressure side channels extends between a pressure side inlet and a pressure side outlet and along a pressure side centerline on a first side of the camber axis, and wherein each of the plurality of suction side channels extends between a suction side inlet and a suction side outlet and along a suction side centerline on a second side of the camber axis.

[0110] The airfoil as in any preceding clause, wherein the pressure side centerline converges towards the camber axis between the pressure side inlet and the pressure side outlet, and wherein the suction side centerline converges towards the camber axis between the suction side inlet and the suction side outlet.

[0111] The airfoil as in any preceding clause, wherein the plurality of pressure side channels converges in cross-sectional area between the pressure side inlet and the pressure side outlet, and wherein the plurality of suction side channels converges in cross-sectional area between the suction side inlet and the suction side outlet.

[0112] The airfoil as in any preceding clause, wherein the trussed structure formed by the plurality of pressure side channels and the plurality of suction side channels is at least partially defined by shared walls of the plurality of pressure side channels and the plurality of suction side channels; and wherein the shared walls extend from the pressure side wall to the suction side wall.

[0113] The airfoil as in any preceding clause, wherein the plurality of pressure side channels and the plurality of suction side channels are radially spaced apart from one another.

[0114] The airfoil as in any preceding clause, wherein each of the plurality of pressure side channels and each of the plurality of suction side channels extend to a downstream plenum, the downstream plenum being defined between outlets of the plurality of pressure side channels and the plurality of suction side channels and the trailing edge.

[0115] The airfoil as in any preceding clause, wherein one or more exit channels extends from the downstream plenum to the trailing edge.

[0116] The airfoil as in any preceding clause, wherein at least one of the plurality of pressure side channels extends to a pressure side exit channel.

[0117] The airfoil as in any preceding clause, wherein the pressure side exit channel is a first pressure side exit channel, and wherein at least one of the plurality of suction side channels includes a second pressure side exit channel disposed downstream of the first pressure side exit channel.

[0118] The airfoil as in any preceding clause, wherein each of the plurality of pressure side channels and each of the plurality of suction side channels define one of a triangle-shaped or oval-shaped cross-sectional area.

[0119] The airfoil as in any preceding clause, wherein the plurality of suction side channels is radially staggered relative to the plurality of pressure side channels such that each suction side channel radially overlaps with at least one pressure side channel of the plurality of pressure side

channels.
[0120] The airfoil as in any preceding clause, wherein the airfoil further defines a dead space disposed between the plurality of pressure side channels and the plurality of suction side channels.
[0121] The airfoil as in any preceding clause, wherein an area of one of the plurality of pressure side channels or the plurality of suction side channels is larger than an area of the other of the plurality of pressure side channels or the plurality of suction side channels.
[0122] A turbomachine comprising: a compressor section; a combustion section disposed downstream of the compressor section; and a turbine section disposed downstream of the combustion section, wherein a rotor blade is disposed in one of the compressor section or the turbine section, the rotor blade including a shank portion and an airfoil extending from the shank portion, the airfoil comprising: a leading edge, a trailing edge, a pressure side wall extending between the leading edge and the trailing edge, and a suction side wall extending between the leading edge and the trailing edge, the pressure side wall and the suction side wall define a main body region and a trailing edge region, the main body region extending from the leading edge to the trailing edge region and the trailing edge region extending from the main body region to the trailing edge; wherein the main body region defines an upstream plenum extending along a span of the airfoil; and wherein the trailing edge region defines a plurality of pressure side channels and a plurality of suction side channels in fluid communication with the upstream plenum and arranged in a trussed structure; wherein the plurality of pressure side channels is disposed adjacent to the pressure side wall, and the plurality of suction side channels is disposed adjacent to the suction side wall.
[0123] The turbomachine as in any preceding clause, wherein: a camber axis is defined halfway between the pressure side wall and the suction side wall; the plurality of pressure side channels is disposed at least partially on a first side of the camber axis and the plurality of suction side channels is disposed at least partially on a second side of the camber axis; and the pressure plurality of side channels and the plurality of suction side channels each converge towards the trailing edge as the plurality of pressure side channels and the plurality of suction side channels extend towards the trailing edge.
[0124] The turbomachine as in any preceding clause, wherein each of the plurality of pressure side channels extends between a pressure side inlet and a pressure side outlet and along a pressure side centerline on a first side of the camber axis, and wherein each of the plurality of suction side channels extends between a suction side inlet and a suction side outlet and along a suction side centerline on a second side of the camber axis.
[0125] The turbomachine as in any preceding clause, wherein the pressure side centerline converges towards the camber axis between the pressure side inlet and the pressure side outlet, and wherein the suction side centerline converges towards the camber axis between the suction side inlet and the suction side outlet.
[0126] The turbomachine as in any preceding clause, wherein the trussed structure formed by the plurality of pressure side channels and the plurality of suction side channels are partially defined by shared walls of the plurality of pressure side channels and the plurality of suction side channels; and wherein the shared walls extend from the pressure side wall to the suction side wall.

## Claims

**1**. An airfoil comprising: a leading edge; a trailing edge; a pressure side wall extending between the leading edge and the trailing edge; and a suction side wall extending between the leading edge and the trailing edge; wherein the pressure side wall and the suction side wall define a main body region and a trailing edge region, the main body region extending from the leading edge to the trailing edge region and the trailing edge region extending from the main body region to the trailing edge; wherein the main body region defines an upstream plenum extending along a span of the airfoil; and wherein the trailing edge region defines a plurality of pressure side channels and a plurality of suction side channels in fluid communication with the upstream plenum and arranged in a trussed structure; and wherein the plurality of pressure side channels is disposed adjacent to the pressure side wall, and the plurality of suction side channels is disposed adjacent to the suction side wall.

**2**. The airfoil as in claim 1, wherein: a camber axis is defined halfway between the pressure side wall and the suction side wall; the plurality of pressure side channels is disposed at least partially on a first side of the camber axis and the plurality of suction side channels is disposed at least partially on a second side of the camber axis; and the plurality of pressure side channels and the plurality of suction side channels each converge towards the trailing edge as the plurality of pressure side channels and the plurality of suction side channels extend towards the trailing edge.

**3**. The airfoil as in claim 2, wherein each of the plurality of pressure side channels extends between a pressure side inlet and a pressure side outlet and along a pressure side centerline on a first side of the camber axis, and wherein each of the plurality of suction side channels extends between a suction side inlet and a suction side outlet and along a suction side centerline on a second side of the camber axis.

**4**. The airfoil as in claim 3, wherein the pressure side centerline converges towards the camber axis between the pressure side inlet and the pressure side outlet, and wherein the suction side centerline converges towards the camber axis between the suction side inlet and the suction side outlet.

**5**. The airfoil as in claim 4, wherein the plurality of pressure side channels converges in cross-sectional area between the pressure side inlet and the pressure side outlet, and wherein the plurality of suction side channels converges in cross-sectional area between the suction side inlet and the suction side outlet.

**6**. The airfoil as in claim 1, wherein the trussed structure formed by the plurality of pressure side channels and the plurality of suction side channels is at least partially defined by shared walls of the plurality of pressure side channels and the plurality of suction side channels; and wherein the shared walls extend from the pressure side wall to the suction side wall.

**7**. The airfoil as in claim 1, wherein the plurality of pressure side channels and the plurality of suction side channels are radially spaced apart from one another.

**8**. The airfoil as in claim 1, wherein each of the plurality of pressure side channels and each of the plurality of suction side channels extends to a downstream plenum, the downstream plenum being defined between outlets of the plurality of pressure side channels and the plurality of suction side channels and the trailing edge.

**9**. The airfoil as in claim 8, wherein one or more exit channels extends from the downstream plenum to the trailing edge.

**10**. The airfoil as in claim 1, wherein at least one of the plurality of pressure side channels extends to a pressure side exit channel.

**11**. The airfoil as in claim 10, wherein the pressure side exit channel is a first pressure side exit channel, and wherein at least one of the plurality of suction side channels includes a second pressure side exit channel disposed downstream of the first pressure side exit channel.

**12**. The airfoil as in claim 1, wherein each of the plurality of pressure side channels and each of the plurality of suction side channels define one of a triangle-shaped or oval-shaped cross-sectional area.

**13**. The airfoil as in claim 1, wherein the plurality of suction side channels is radially staggered relative to the plurality of pressure side channels such that each suction side channel radially overlaps with at least one pressure side channel of the plurality of pressure side channels.

**14**. The airfoil as in claim 1, wherein the airfoil further defines a dead space disposed between the plurality of pressure side channels and the plurality of suction side channels.

**15**. The airfoil as in claim 1, wherein an area of one of the plurality of pressure side channels or the plurality of suction side channels is larger than an area of the other of the plurality of pressure side channels or the plurality of suction side channels.

**16**. A turbomachine comprising: a compressor section; a combustion section disposed downstream of the compressor section; and a turbine section disposed downstream of the combustion section, wherein a rotor blade is disposed in one of the compressor section or the turbine section, the rotor blade including a shank portion and an airfoil extending from the shank portion, the airfoil comprising: a leading edge, a trailing edge, a pressure side wall extending between the leading edge and the trailing edge, and a suction side wall extending between the leading edge and the trailing edge, the pressure side wall and the suction side wall define a main body region and a trailing edge region, the main body region extending from the leading edge to the trailing edge region and the trailing edge region extending from the main body region to the trailing edge; wherein the main body region defines an upstream plenum extending along a span of the airfoil; and wherein the trailing edge region defines a plurality of pressure side channels and a plurality of suction side channels in fluid communication with the upstream plenum and arranged in a trussed structure; and wherein the plurality of pressure side channels is disposed adjacent to the pressure side wall, and the plurality of suction side channels is disposed adjacent to the suction side wall.

**17**. The turbomachine as in claim 16, wherein: a camber axis is defined halfway between the pressure side wall and the suction side wall; the plurality of pressure side channels is disposed at least partially on a first side of the camber axis and the plurality of suction side channels is disposed at least partially on a second side of the camber axis; and the pressure plurality of side channels and the plurality of suction side channels each converge towards the trailing edge as the plurality of pressure side channels and the plurality of suction side channels extend towards the trailing edge.

**18**. The turbomachine as in claim 17, wherein each of the plurality of pressure side channels extends between a pressure side inlet and a pressure side outlet and along a pressure side centerline on a first side of the camber axis, and wherein each of the plurality of suction side channels extends between a suction side inlet and a suction side outlet and along a suction side centerline on a second side of the camber axis.

**19**. The turbomachine as in claim 18, wherein the pressure side centerline converges towards the camber axis between the pressure side inlet and the pressure side outlet, and wherein the suction side centerline converges towards the camber axis between the suction side inlet and the suction side outlet.

**20**. The turbomachine as in claim 16, wherein the trussed structure formed by the plurality of pressure side channels and the plurality of suction side channels are partially defined by shared walls of the plurality of pressure side channels and the plurality of suction side channels; and wherein the shared walls extend from the pressure side wall to the suction side wall.